Topic 2.4 Simple and giant structures

Exercise 2.4A Ionic or covalent bonds

1 Ionic bonding, because the particles are packed closely together in a lattice pattern.

2 Arrows should indicate the points where the large atom and the four small atoms in each molecule join. The label should read 'strong forces within each molecule'.

3 Substance B

4 Substance A

5 Ionic

6 covalent

7 Any correct example, such as graphite or silicon dioxide.

Exercise 2.4B Properties of ionic and covalent substances

1 Its melting and boiling points are high.

2 a gas

3 potassium chloride, calcium chloride and magnesium oxide

4 ammonia and bromine

5 Water; because it has a boiling point of 100 °C and a melting point of 0 °C.

6 bromine

7 Magnesium oxide has high melting and boiling points because it is an ionic compound. The ions have strong forces holding them together in a lattice pattern. This means that a lot of energy is needed to overcome these forces and make the magnesium oxide melt or boil.

8 Ammonia has low melting and boiling points because, although the forces inside the molecules are strong, the forces between the molecules are weak and less energy is needed to overcome these intermolecular forces and make the ammonia melt or boil.

Exercise 2.4C Giant structures of carbon

1 diamond (left) and graphite (right)

2 The atoms of carbon in graphite are arranged in layers. The bonds between the carbon atoms in the layers are strong (each carbon atom bonds with three other atoms) but the bonds between the layers are weak. This means that the layers can slide over one another and the surface is soft and comes away. For example, when you write with a graphite pencil.

3 The atoms of carbon in diamond are arranged in a rigid, giant three-dimensional structure or lattice. This means that there are strong bonds throughout the whole structure and this is what makes diamond so hard.